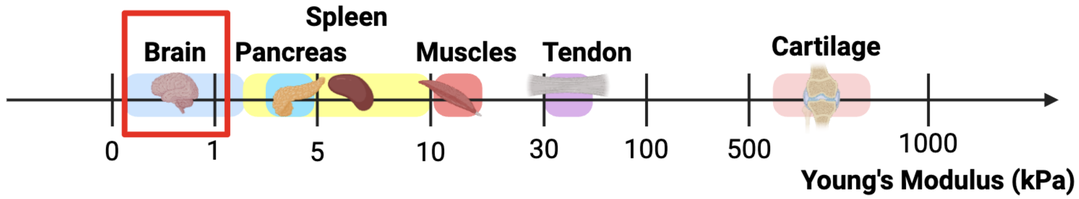
Production of Extracellular Vesicles from Encapsulated Cells in Zwitterionic Hydrogels

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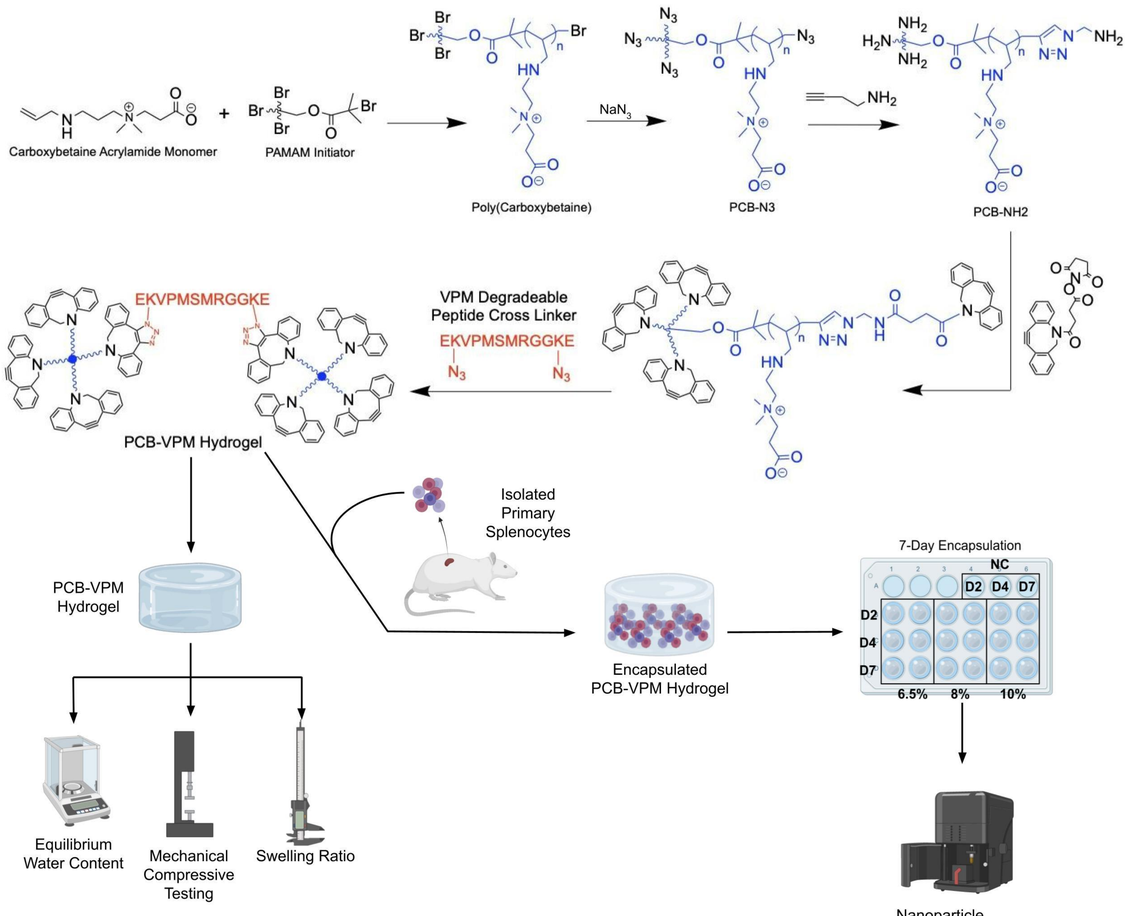
INTRODUCTION

- Extracellular vesicles (EVs) are naturally secreted from cells and are promising drug delivery vehicles due to their biocompatibility and targeting abilities [1].
- Intravenous delivery of EVs often fails to achieve sustained release or maintain therapeutic concentrations over time [2].
- This study explores a PCB hydrogel culture system to support and preserve EV-secreting cells for superior EV production in the brain.



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METHODS



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RESULTS

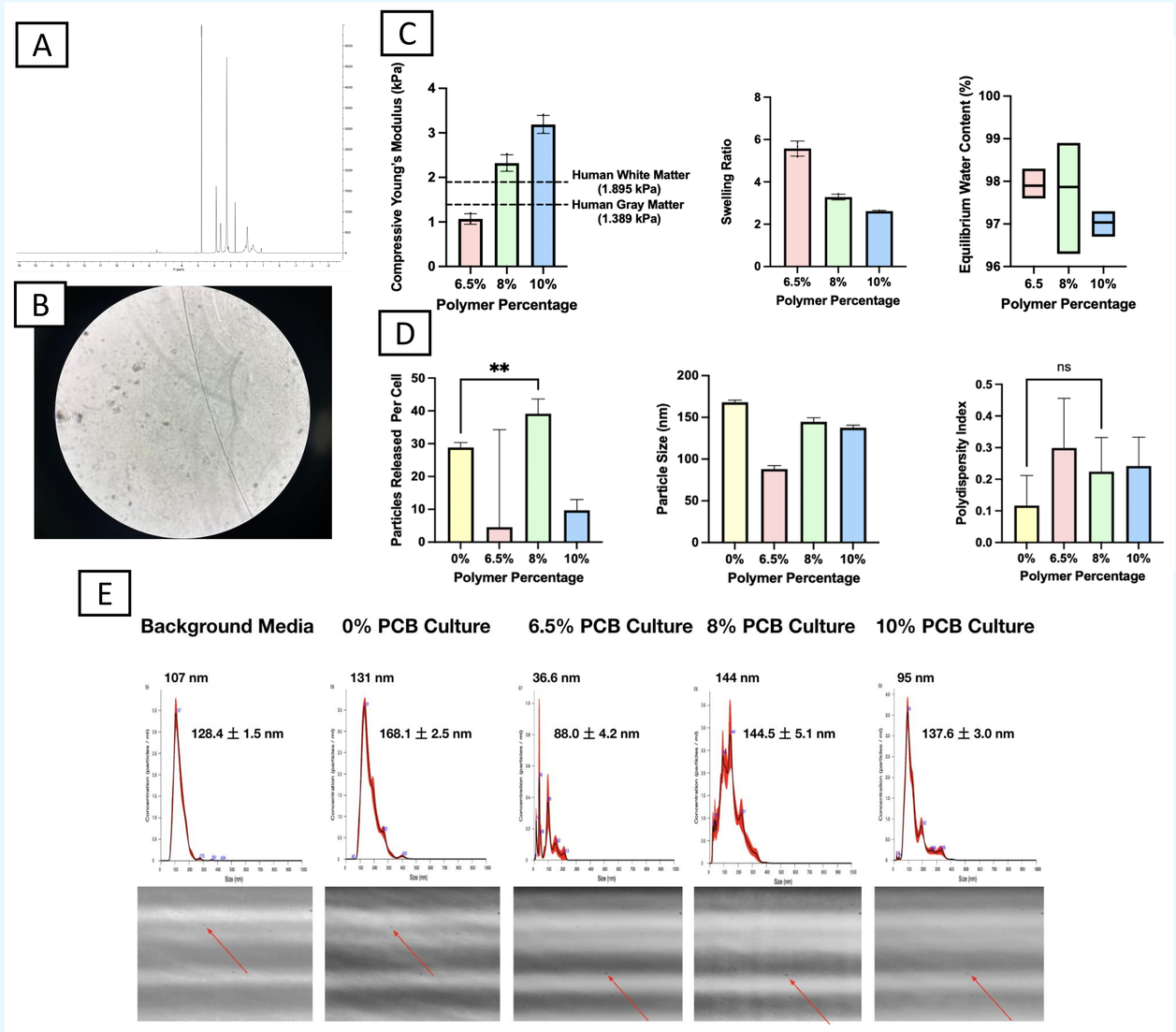


Figure 1. A. The H1-NMR spectrum of the 35kDa PCB-DBCO. B. Image of the 6.5% encapsulated culture. C. Mechanical characterization of the PCB-VPM hydrogels. D. Released particle characterization. E. NTA particle distributions and image of released particles.

CONCLUSIONS

- Although 6.5% PCB-VPM better recapitulates the brain mechanical environment, 8% showed the highest particle release at an acceptable particle size and PDI.
- Future steps include the identification of the release particles and increasing the seeding density.

REFERENCES

[1] Du et al., Journal of Nanobiotechnology, 2023.
[2] Driedonks et al., J Extracell Biol., 2022.

ACKNOWLEDGEMENTS

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